Independent Learning in Constrained Markov Potential Games

Philip Jordan **Anas Barakat** **Niao He**

ETH Zürich

1 Independent Learning in 3 Constrained Markov 2 Potential Games

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Outline

1. Preliminaries: Markov games, plus . . .
   1. independent learning
   2. potential structure (MPGs)
   3. constraints (CMPGs)

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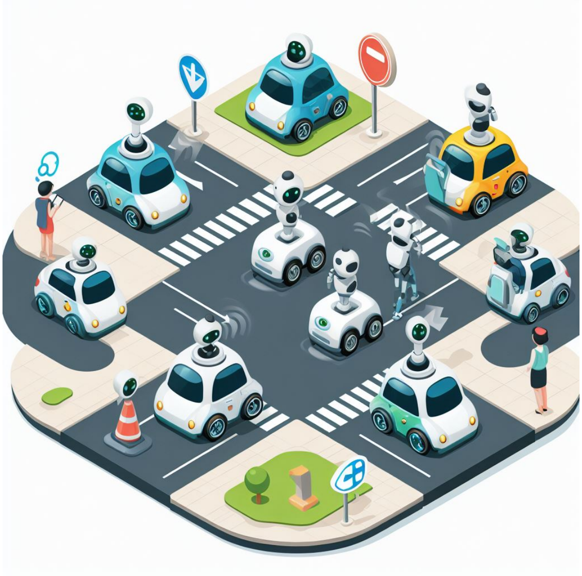

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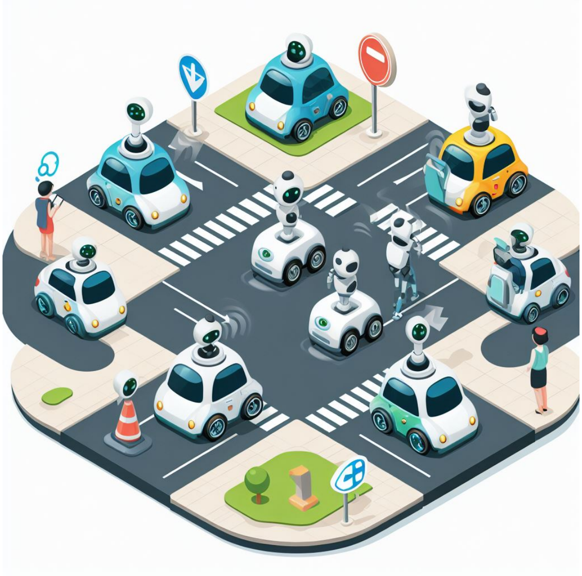
Markov Game (MG):



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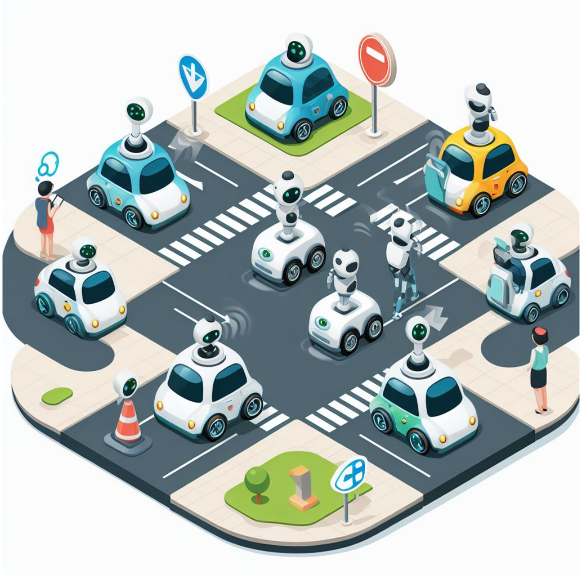
- players $\mathcal{N} = \{1, \ldots, m\}$



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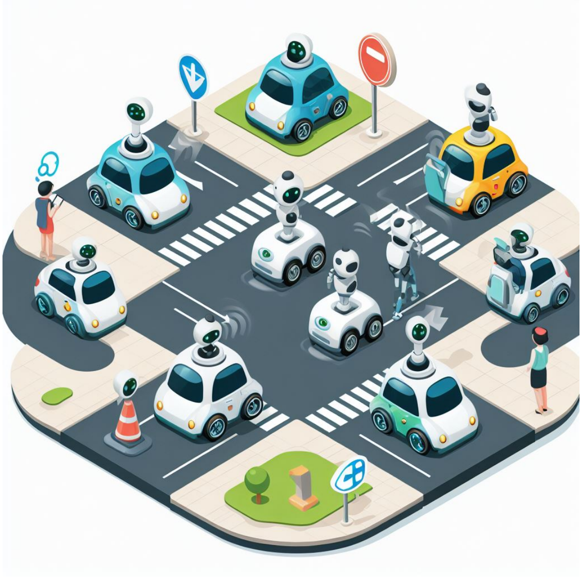

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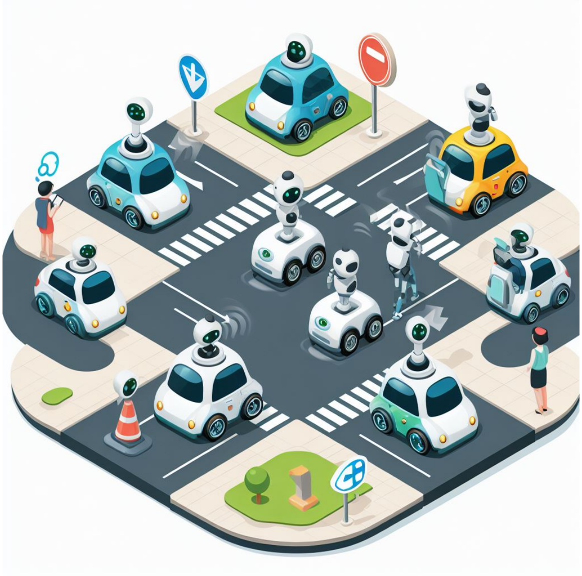

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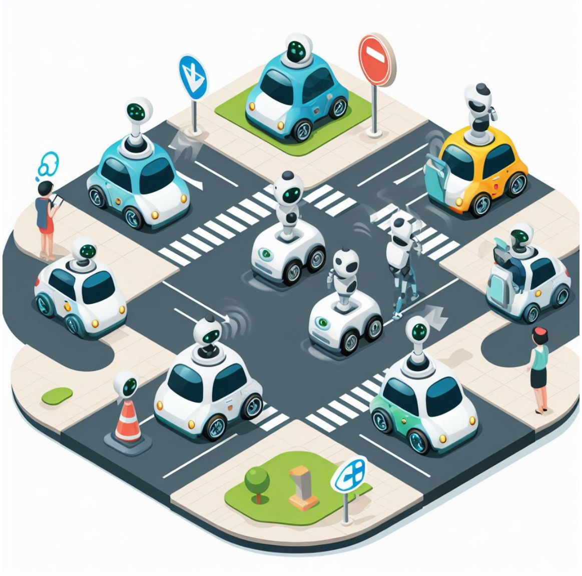

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For **joint** policy $\pi \in \Pi = \prod_{i \in \mathcal{N}} \Delta(\mathcal{A}_i)^{\mathcal{S}}$,

$$V_{r_i}(\pi) := \mathbb{E}_{s \sim \mu} \left[\sum_{t=0}^{T} r_i(s_t, a_t) \mid s_0 = s \right]$$

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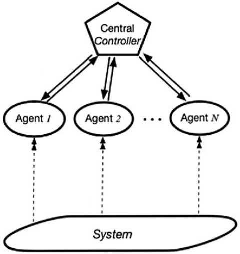
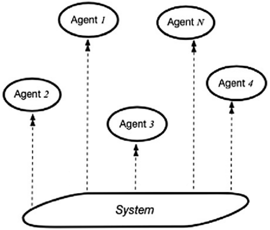
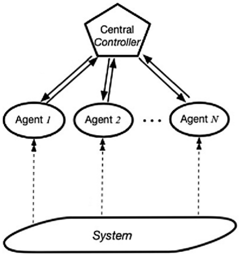


centralized



1 independent learning:
(no coordination among players)



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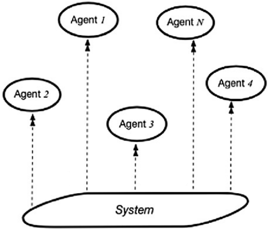

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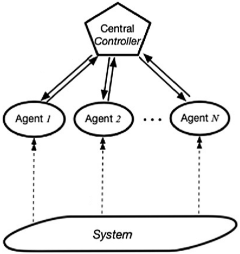

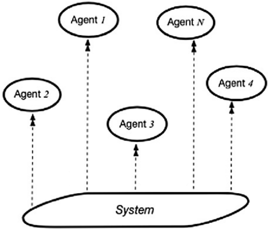

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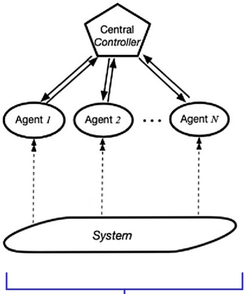

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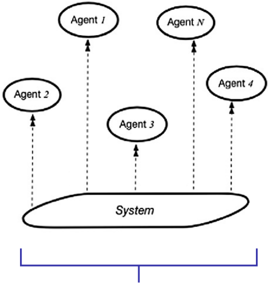

1 independent learning:
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- communication cost/latency
- competitive environments (e.g. markets)
- scaling to large number of players

Why ② potential structure (MPGs)?

- **ϵ-approximate Nash equilibrium (ϵ-NE):** $\pi^* \in \Pi$ s.t. $\forall i \in \mathcal{N}, \pi_i' \in \Pi^i$,

$$V_{r_i}(\pi^*) - V_{r_i}(\pi_i', \pi_{-i}^*) \leq \epsilon.$$

Why (2) potential structure (MPGs)?

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- finding ϵ-NE of MG is generally not tractable [Rub16, Das13]
- **Markov Potential Game (MPG)**:

  $\exists \Phi : \Pi \to \mathbb{R}$ s.t. $\forall i \in \mathcal{N}$, $(\pi_i, \pi_{-i}) \in \Pi$, and $\pi_i' \in \Pi_i'$,
$$V_{r_i}(\pi_i, \pi_{-i}) - V_{r_i}(\pi_i', \pi_{-i}) = \Phi(\pi_i, \pi_{-i}) - \Phi(\pi_i', \pi_{-i})$$

③ constraints (CMPGs)

- subset of feasible policies $\Pi_c := \{\pi \in \Pi \mid V_c(\pi) \leq \alpha\}$ for some $\alpha \in \mathbb{R}$ where

$$V_c(\pi) := \mathbb{E}_{s \sim \mu}\left[\sum_{t=0}^{T} c(s_t, a_t) \mid s_0 = s\right]$$

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MPG	Nash-CA [SMB22]	Independent PGA [LOPP22],[ZRL22],[DWZJ22]
CMPG	CA-CMPG [ARHK23]	**?**

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Related work (unconstrained)

Nash-CA [SMB22]

- turn-based for $i \in \mathcal{N}$; fix $\pi_{-i}^{(t)}$,

$$\hat{\pi}_i^{(t+1)} = \arg\max_{\pi_i \in \Pi^i} V_{r_i}(\pi_i, \pi_{-i}^{(t)})$$

(by solving single-agent MDP)

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- Either
  - $\pi^{(t)}$ is ϵ-NE, or
  - $\exists i \in \mathcal{N}$ s.t.

$$V_{r_i}(\pi_i^{(t)}, \pi_{-i}^{(t)}) - V_{r_i}(\hat{\pi}_i^{(t+1)}, \pi_{-i}^{(t)}) > \epsilon$$

i.e. can still decrease Φ by $> \epsilon$

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- simultaneously $\forall i \in \mathcal{N}$,

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What changes for CMPGs?

CA-CMPG [ARHK23]

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(by solving single-agent CMDP)

Independent learning in CMPGs

?

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- strong duality *does not* hold [ARHK23]

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Our Approach: Idea

$$\min_{(\pi_1,\ldots,\pi_m)\in\Pi_c} \Phi(\pi) \tag{1}$$

where $\Pi_c := \{\pi \in \Pi \mid V_c(\pi) \geq \alpha\}$

Lemma

If π is an ϵ-KKT policy of (1), then π is a constrained $\mathcal{O}(\epsilon)$-NE.

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- Solve via gradient switching subroutine:

$$\pi^{(t,k+1)} = \begin{cases} \mathcal{P}_\Pi\left[\pi^{(t,k)} - \nu_k \nabla_\pi \Phi_{\eta,\pi^{(t,k)}}\left(\pi^{(t,k)}\right)\right] & \text{if } V^c_{\eta,\pi^{(t,k)}}\left(\pi^{(t,k)}\right) - \alpha \leq \delta_k, \\ \mathcal{P}_\Pi\left[\pi^{(t,k)} - \nu_k \nabla_\pi V^c_{\eta,\pi^{(t,k)}}\left(\pi^{(t,k)}\right)\right] & \text{otherwise} \end{cases}$$

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- independent implementation?

Independent implementation

- **Observation:**

$$\nabla_{\pi_i}\Phi_{\eta,\pi'}(\pi) = \nabla_{\pi_i}\Phi(\pi) + \frac{1}{\eta}\left(\pi_i - \pi_i'\right) = \nabla_{\pi_i} V_{r_i}(\pi) + \frac{1}{\eta}\left(\pi_i - \pi_i'\right)$$

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is equivalent to independently, for all $i \in \mathcal{N}$,

$$\pi_i^{(t,k+1)} = \begin{cases} \mathcal{P}_{\Pi^i}\left[\pi_i^{(t,k)} - \nu_k \nabla_{\pi_i} V_{r_i}(\pi^{(t,k)}) - \frac{\nu_k}{\eta}(\pi_i^{(t,k)} - \pi_i^{(t)})\right] \\ \qquad \text{if } V_c(\pi^{(t,k)}) + \frac{1}{2\eta}\|\pi^{(t,k)} - \pi^{(t)}\|^2 - \alpha \le \delta_k \\ \mathcal{P}_{\Pi^i}\left[\pi_i^{(t,k)} - \nu_k \nabla_{\pi_i} V_c(\pi^{(t,k)}) - \frac{\nu_k}{\eta}(\pi_i^{(t,k)} - \pi_i^{(t)})\right] \\ \qquad \text{otherwise} \end{cases}$$

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Algorithm 1 iProxCMPG: independent **Prox**imal-policy algorithm for **CMPG**s

initialization: $\pi^{(0)} \in \Pi^{\xi}$ s.t. $V_c(\pi^{(0)}) < \alpha$ and suitably chosen $\eta, \xi, T, K, \{(\nu_k, \delta_k)\}_{0 \le k \le K}$

for $t = 0, \ldots, T-1$ **do**

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  for $k = 0, \ldots, K-1$ and $i \in \mathcal{N}$ simultaneously **do**

    sample B trajectories by following $\pi_i^{(t,k)}$ to estimate $\hat{V}_c(\pi^{(t,k)}), \hat{\nabla} V_{\pi_i}^{r_i}(\pi^{(t,k)}), \hat{\nabla} V_{\pi_i}^{c}(\pi^{(t,k)})$

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  $\pi_i^{(t+1)} = \pi_i^{(t,\hat{k})}$ where $\hat{k}$ is sampled from $\{k \in [K] \mid \hat{V}_c(\pi^{(t,k)}) \le \delta_k\}$

output: $\pi_i^{(T)}$ for $i \in \mathcal{N}$

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Comparison

	centralized	**independent**
MPG	Nash-CA [SMB22] $\mathcal{O}(\epsilon^{-3})$	Independent PGA [LOPP22],[ZRL22],[DWZJ22] $\mathcal{O}(\epsilon^{-5})$
CMPG	CA-CMPG [ARHK23] $\tilde{\mathcal{O}}(\epsilon^{-5})$	*Our algorithm* $\tilde{\mathcal{O}}(\epsilon^{-7})$

Outline

1. Preliminaries: Markov games, plus ...
   1. independent learning
   2. potential structure (MPGs)
   3. constraints (CMPGs)
2. Related work & challenges
3. Our approach
   a) Idea
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Simulations

- **Pollution tax model**
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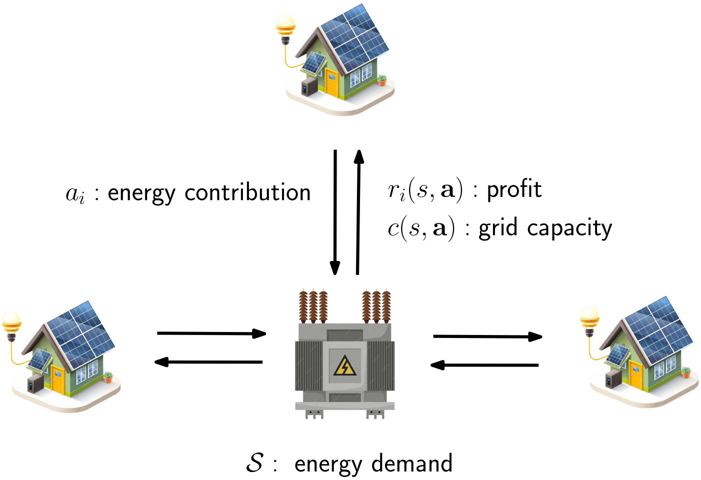

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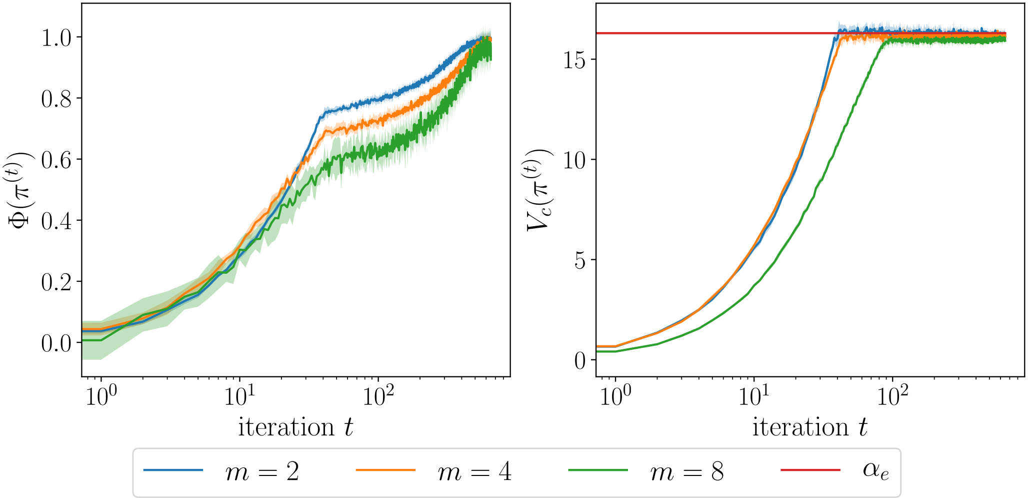

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Future Directions

- Are **sample complexity gaps** inherent?

	centralized	**independent**
MPG	Nash-CA [SMB22] $\mathcal{O}(\epsilon^{-3})$	Independent PGA [LOPP22],[ZRL22],[DWZJ22] $\mathcal{O}(\epsilon^{-5})$
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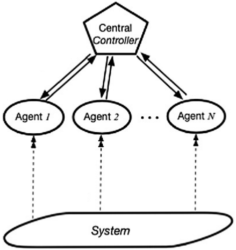
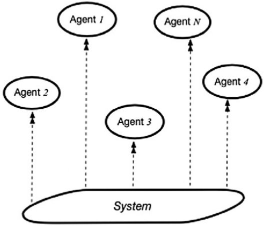
Future Directions

- Any **provable** benefit of centralization?



centralized



independent learning

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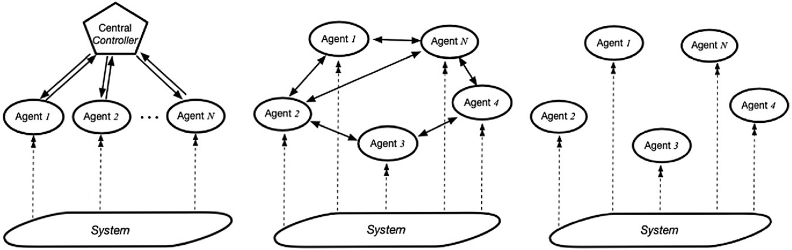

centralized

networked agents

independent learning

- dependence on graph topology
- robustness against malicious agents

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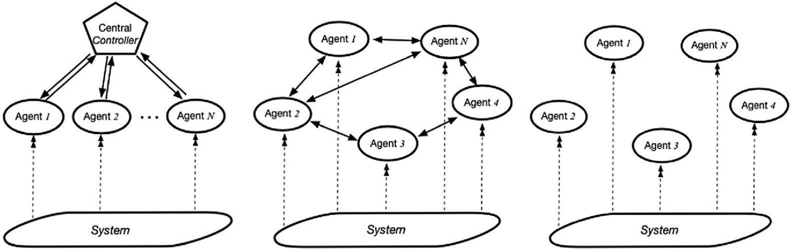

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independent learning

"fully" independent learning

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Appendix

Simulations

Distributed energy marketplace

- m energy providers, choosing amount of energy to contribute $\mathcal{A}_i = \{0, \ldots, A_i - 1\}$

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- $\mathcal{S} = \{0, \ldots, S - 1\}$ energy demand (high to low)
- profit $r_i(s, a_i, a_{-i}) = c_0 a_i^2 - c_1 a_i^2 \sum_{i \in \mathcal{N}} a_i - a_i c_2^s$ for some $c_0, c_1, c_2 \in \mathbb{R}$

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- sample $w \sim \mathcal{U}(\{0, 1, \ldots, W\})$ and set

$$s' = \begin{cases} \mathcal{P}_{[0,S-1]}\left(\sum_{i \in \mathcal{N}} a_i - w\right) & \text{w.p. } 0.9 \\ w & \text{w.p. } 0.1 \end{cases}$$

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- $c(s, \mathbf{a}) = \sum_{i \in \mathcal{N}} a_i$, require $V_c(\pi) \leq \alpha_e$ for some $\alpha_e \in \mathbb{R}$

$\implies$ satisfies CMPG condition, see [NLKS22]

Simulations

Pollution tax model

- m factories that choose production volume $\mathcal{A}_i = \{L, H\}$

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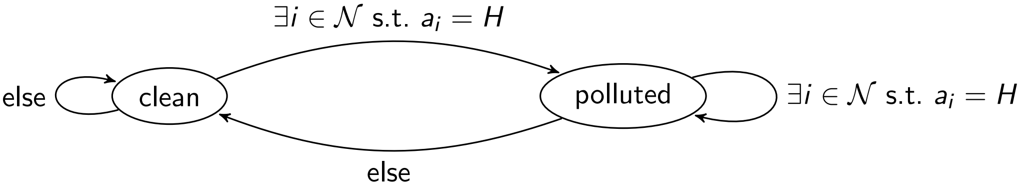

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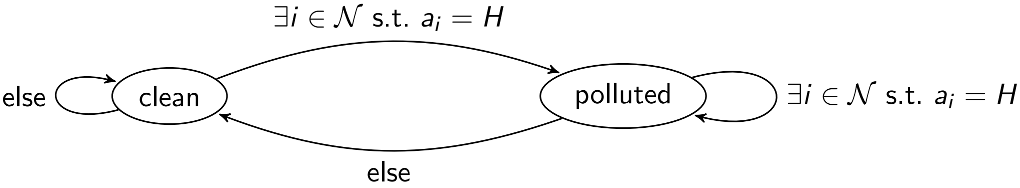
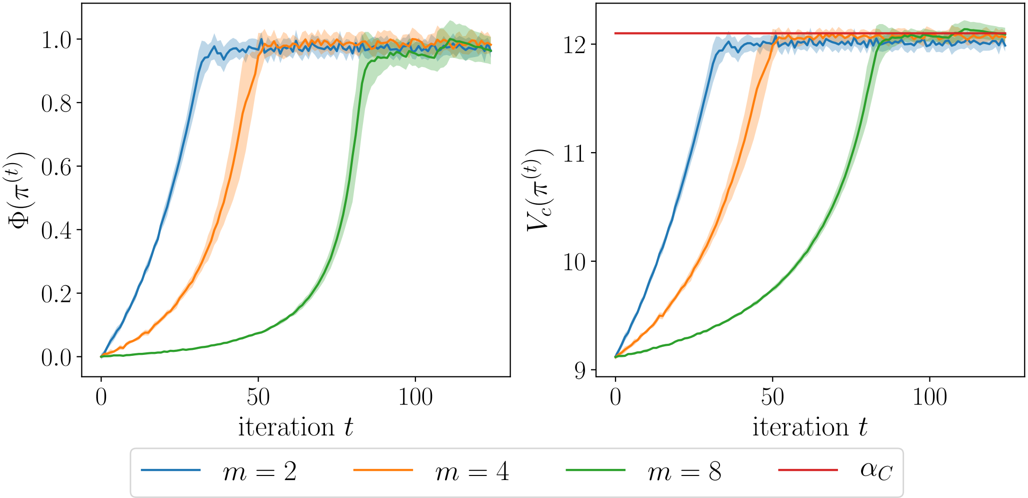


-
- let $r_i(s, \mathbf{a}) = -T_P \mathbb{I}_{\{s=\text{polluted}\}} + \begin{cases} P_L & \text{if } a_i = L \\ P_H & \text{else} \end{cases}$ and $c(s, \mathbf{a}) = |\{i \in \mathcal{N} \mid a_i = H\}|$

Simulations



Iteration & Sample Complexity Result

Assumptions

- **Initial feasibility:** $\pi^{(0)}$ *satisfies* $V_c(\pi^{(0)}) < \alpha$
- **Uniform Slater's condition:**

$$\exists \zeta > 0 \text{ s.t. } \forall \pi' \in \Pi \text{ with } V_c(\pi') < \alpha, \exists \pi \in \Pi \text{ s.t. } V^c_{\eta,\pi'}(\pi) \leq \alpha - \zeta$$

Theorem

For $\epsilon > 0$*, using* ϵ*-greedy exploration, after running iProxCMPG for suitably chosen* $\eta, T, K,$ *and* $\{(\nu_k, \delta_k)\}_{0 \leq k \leq K}$*, there exists* $t \in [T]$ *s.t. in expectation* $\pi^{(t)}$ *is a constrained* ϵ*-NE.*

- **Exact gradients:** *total iteration complexity*[1] $\tilde{\mathcal{O}}(\epsilon^{-4})$
- **Finite sample:** *total sample complexity*[1] $\tilde{\mathcal{O}}(\epsilon^{-7})$

[1] $\tilde{\mathcal{O}}(\cdot)$ hides logarithmic dependencies in $1/\epsilon$, and polynomial dependencies in $m, S, A_{\max}, 1-\gamma, \zeta$, and D.

Proof idea (exact gradients).

1. for $K = \mathcal{O}(\epsilon^{-2})$, inner loop guarantees sufficiently exact proximal update
2. for $T = \mathcal{O}(\epsilon^{-2})$, outer loop guarantees $\exists t \in [T]$ s.t. $\left\|\pi^{(t+1)} - \pi^{(t)}\right\| = \mathcal{O}(\epsilon)$

$\implies$ $\pi^{(t+1)}$ satisfies ϵ-KKT conditions for $\min_{\pi \in \Pi_c} \Phi(\pi)$

$\implies$ $\pi^{(t+1)}$ satisfies ϵ-KKT conditions for playerwise problem with $\pi_{-i}^{(t+1)}$ fixed:

$$\min_{\pi_i \in \Pi_c^i(\pi_{-i}^{(t+1)})} V_{r_i}(\pi_i, \pi_{-i}^{(t+1)}) \tag{2}$$

$\overset{gr.dom.}{\implies}$ for all $i \in \mathcal{N}$, bound duality gap of (2) via gradient dominance

$\implies$ together with $V_c(\pi^{(t+1)}) \leq \alpha$, it follows that $\pi^{(t+1)}$ is constrained $\mathcal{O}(\epsilon)$-NE

$\square$

Proof idea (finite sample).

Algorithm 1 iProxCMPG: **i**ndependent **Prox**imal-policy algorithm for **CMPG**s $\mathcal{O}(\epsilon^{-7})$

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for $t = 0, \ldots, T-1$ **do** $\leftarrow$ $\mathcal{O}(\epsilon^{-2})$ times

$\quad \pi_i^{(t,0)} = \pi_i^{(t)}$

$\quad$ **for** $k = 0, \ldots, K-1$ and $i \in \mathcal{N}$ simultaneously **do** $\leftarrow$ $\mathcal{O}(\epsilon^{-3})$ times

$\quad\quad$ sample B trajectories by following $\pi_i^{(t,k)}$ to estimate $\hat{V}_{r_i}(\pi^{(t,k)}), \underline{\hat{\nabla} V_{\pi_i}^{r_i}(\pi^{(t,k)}), \hat{\nabla} V_{\pi_i}^{c}(\pi^{(t,k)})}$

$\quad\quad$ $B = \mathcal{O}(\epsilon^{-2})$

$$\pi_i^{(t,k+1)} = \begin{cases} \mathcal{P}_{\Pi^{i,\xi}}\left[\pi_i^{(t,k)} - \nu_k \hat{\nabla}_{\pi_i} V_{r_i}(\pi^{(t,k)}) - \frac{\nu_k}{\eta}(\pi_i^{(t,k)} - \pi_i^{(t)})\right] & \text{if } \hat{V}_c(\pi^{(t,k)}) - \alpha \le \delta_k \\ \mathcal{P}_{\Pi^{i,\xi}}\left[\pi_i^{(t,k)} - \nu_k \hat{\nabla}_{\pi_i} V_c(\pi^{(t,k)}) - \frac{\nu_k}{\eta}(\pi_i^{(t,k)} - \pi_i^{(t)})\right] & \text{otherwise} \end{cases}$$

variance $\mathcal{O}(\epsilon^{-1})$

$\quad \pi_i^{(t+1)} = \pi_i^{(t,\hat{k})}$ where $\hat{k}$ is sampled from $\{k \in [K] \mid \hat{V}_c(\pi^{(t,k)}) \le \delta_k\}$

output: $\pi_i^{(T)}$ for $i \in \mathcal{N}$

□